

ENTITY ANALYSIS

CHAPTER-2

2.1 INTRODUCTION:

This chapter focuses on the analysis and design of the **Amazon-Inspired E-Commerce Database Management System**. A well-designed database is the foundation of every successful e-commerce platform because it stores, organizes, and manages large volumes of data related to customers, products, orders, payments, and deliveries. Without an efficient database, it becomes difficult to maintain data accuracy, process transactions quickly, and provide a smooth shopping experience to users.

The primary objective of this chapter is to identify and analyze the major components required for designing the database system. The analysis begins by identifying the **entities** that represent the real-world objects of the e-commerce platform, such as **Customer, Seller, Product, Category, Orders, Payment, Shipment, Cart, Inventory, and Review**. Each entity contains a set of **attributes** that store detailed information related to that entity. For example, a Customer entity stores details such as name, email, phone number, and address, while a Product entity stores information such as product name, description, brand, and price.

In addition, **Primary Keys (PK)** are identified to uniquely distinguish each record within an entity, and **Foreign Keys (FK)** are identified to establish relationships between different entities. The relationships among entities define how data is connected and exchanged within the system. Examples include a customer placing multiple orders, a seller selling multiple products, and an order having associated payment and shipment details.

Furthermore, this chapter identifies the **cardinality and participation constraints** of relationships to ensure proper representation of real-world business operations. The identification of entities, attributes, keys, and relationships provides the necessary foundation for developing the **Entity Relationship Diagram (ER Diagram)** and implementing a **normalized**,

scalable, secure, and efficient relational database for the Amazon-inspired e-commerce platform.

ENTITY-CUSTOMER

The Customer entity stores the information of users who register and shop on the e-commerce platform. It maintains customer details such as personal information, contact details, and address information. This entity helps in managing customer accounts, orders, and shopping activities.

Attribute	Key Type
Customer_ID	Primary Key
First_Name	Not null
Last name	Not null
Email	Unique Key
Phone	Unique key
Address	-
City	-
State	-
Pincode	-

ENTITY-SELLER

The Seller entity stores information about vendors who sell products on the platform. It contains seller details such as name, contact information, and

business details. This entity enables sellers to manage product listings and inventory.

Attribute	Key Type
Seller_ID	Primary Key
Seller_Name	Not null
Email	Unique Key
Phone	Unique key
Business_Name	Not Null
Address	-

ENTITY-CATEGORY

The Category entity is used to organize products into different groups such as Electronics, Fashion, and Books. Categorizing products helps customers browse and search products efficiently. It improves product management and navigation within the platform.

Attribute	Key Type
Category_ID	Primary Key
Category_Name	Unique key
Description	-

ENTITY-PRODUCT

The Product entity stores information about items available for sale on the platform. It contains details such as product name, description, brand, and price. This entity acts as the core component of the e-commerce system because all shopping activities revolve around products.

Attribute	Key Type
Product_ID	Primary Key
Product_Name	Not Null
Description	-
Brand	-

ENTITY-INVENTORY

The Inventory entity maintains the stock information of products available on the platform. It records the quantity of products and tracks inventory updates. This entity helps prevent stock shortages and supports efficient inventory management.

Attribute	Key Type
Inventory_ID	Primary Key
Product_ID	Foreign Key
Stock_Quantity	Not Null
Last_Updated	-

ENTITY-CART

The Cart entity represents the shopping cart of a customer. It stores products that customers intend to purchase before placing an order. This entity provides flexibility by allowing customers to add, remove, or update items before checkout.

Attribute	Key Type
Cart_ID	Primary Key
Customer_ID	Foreign Key
Created_Date	-

ENTITY-ORDERS

The Orders entity stores information about the orders placed by customers. It records details such as order date, order status, and total amount. This entity is essential for tracking purchases and managing order processing.

Attribute	Key Type
Order_ID	Primary Key
Customer_ID	Foreign Key
Order_Date	Not Null
Total_Amount	Not Null
Order_Status	-

The Orders entity stores information about the orders placed by customers. It records details such as order date, order status, and total amount. This entity is essential for tracking purchases and managing order processing.

ENTITY-PAYMENT

The Payment entity stores transaction information related to customer orders. It maintains details such as payment method, amount, payment status, and payment date. This entity ensures secure and accurate payment processing.

Attribute	Key Type
Payment_ID	Primary Key
Order_ID	Foreign Key
Payment_Method	Not Null
Amount	Not Null
Payment_Status	-
Payment_Date	-

ENTITY-SHIPMENT

The Shipment entity manages product delivery information. It stores tracking details, delivery addresses, and shipment status. This entity helps customers and administrators monitor the delivery process.

Attribute	Key Type
Shipment_ID	Primary Key
Order_ID	Foreign Key
Tracking_Number	Unique Key
Delivery_Address	Not Null
Shipment_Status	-
Delivery_Date	-

ENTITY-REVIEW

The Review entity stores customer feedback and ratings for products. It contains information such as ratings, comments, and review dates. This entity helps

customers make informed purchasing decisions and provides valuable feedback to sellers.

Attribute	Key Type
Review_ID	Primary Key
Customer_ID	Foreign Key
Product_ID	Foreign Key
Rating	Not Null
Comment	-
Review_Date	-

2.3 RELATIONSHIPS BETWEEN ENTITIES:

This table represents the relationships that exist among the entities of the Amazon-Inspired E-Commerce Database Management System. It identifies how entities interact with each other and specifies the cardinality of each relationship, such as One-to-One, One-to-Many, and Many-to-Many.

Entity 1	Relationship	Entity 2	Cardinality
Seller	Sells	Product	1 : M
Category	Contains	Product	1 : M
Product	Has	Inventory	1 : 1
Customer	Owns	Cart	1 : 1
Cart	Contains	Product	M : M
Customer	Places	Orders	1 : M

Orders	Has	Payment	1 : 1
Orders	Shipped By	Shipment	1 : 1
Customer	Writes	Review	1 : M
Product	Receives	Review	1 : M

2.4 ENTITIES AND RELATIONSHIPS

This table provides a detailed representation of the relationships among entities, including their cardinalities and participation constraints. It helps in understanding how data is interconnected and ensures proper database design and data integrity.

Entities	Relation	Cardinality	Type of Participation
Customer	Owns	OneToOne	Partial
Customer	Places	OneToMany	Partial
Customer	Writes	OneToMany	Partial
Seller	Sells	OneToMany	Partial
Category	Contains	OneToMany	Total
Product	Has	OneToOne	Total
Product	Receives	OneToMany	Partial
Cart	Contains	ManyToMany	Partial
Orders	Has	OneToOne	Total
Orders	Shipped By	OneToOne	Total